

IVECO AT THE 2012 IAA MOTOR SHOW IN HANOVER

THE NEW STARS OF THE IVECO HEAVY RANGE AT IAA 2012

- The new STRALIS in the spotlight
- STRALIS Emotional: a novelty “super-equipped”
- The new Trakker: a redesigned cab and improved comfort
- Trakker Dakar

THE IVECO STAND

- The focus is on the heavy range
- FPT Industrial innovative powertrain technologies in the heart of the stand
- At the exhibition area also light and medium ranges
- Astra exhibits HD9 in Hanover
- Iveco collective passenger transport solutions on stage
- The outdoor area: test-drive of the electric Daily
- The innovation area: “Iveco Dual Energy” chassis concept
- Aftersales and financing services

IVECO INNOVATION TECHNOLOGIES FOR FUTURE MOBILITY

- “Iveco Dual Energy”: for tomorrow’s light-duty vehicle

THE IVECO RANGE: TECHNICAL INFORMATION

- STRALIS Hi-Way: working on the hi-way
- STRALIS Hi-Way “Emotional”
- STRALIS LNG Natural Power: alternative tractions for the transport business
- New Trakker: efficiency and value even off-road
- Astra HD9 facing the international audience
- Daily: the light vehicle suitable for every kind of work
- Eurocargo: an ideal solution for any challenge
- Passenger transportation: from minibuses to luxury buses

THE NEW STARS OF THE IVECO HEAVY RANGE AT IAA 2012

On the occasion of the **64th International Commercial Vehicle Show (IAA)** in Hanover, from 20 to 27 September 2012, Iveco presents the new stars of its heavy-duty range: **the new STRALIS and the new Trakker.**

The new STRALIS in the spotlight

After its launch on July 1, the new STRALIS debuts before a large international audience, with two important novelties concerning the choice of Euro VI engines offered. The vehicle, which is the Iveco's answer to the needs of today's professional transport, is enhanced by the new powerful **Cursor 9 with 400 HP and 1700 Nm**, a particularly competitive choice for fuel transport applications and the **Cursor 11 with 480 HP and 2250 Nm**, which offers the same performance as a 13-litre engine, combined with significant advantages in terms of weight and fuel consumption.

With the new STRALIS, Iveco reinforces its customer-oriented strategy, **reducing total cost of ownership (TCO) of up to 4% for long haulage missions**. The vehicle is characterised by highly innovative new engine technology, the **patented FPT Industrial system, Hi-eSCR**, which meets the Euro VI regulations without compromising fuel efficiency.

An additional novelty for 2012, displayed in Hanover, is the **STRALIS LNG Natural Power** version, a vehicle powered by liquefied natural gas that is ideally suited to medium-range distribution and night-time delivery applications, which starting this year, has been marketed all over Europe.

STRALIS Emotional: a novelty "super-equipped"

One of the new additions to the heavy vehicle range presented by Iveco is the STRALIS "Emotional", a vehicle offering a huge choice of additional accessories to create a pleasant driving environment and an area of complete relaxation on board. These factors contribute to increased productivity and vehicle safety.

The new Trakker: a redesigned cab and improved comfort

In order to complete the heavy-duty range renewal, Iveco presents, in its world premiere, the **New Trakker**, an off-road construction site vehicle, with an entirely renewed cab. The cab has been conceived to enhance driving comfort and life on board, factors already considered to be fundamental for productivity and safety.

The new Trakker operates on impracticable uneven terrain, relying on its Cursor engines (8 and 13 Euro V), which ensure power and resistance over years for a vehicle that must be operative 365 days per year. The new

Trakker significantly increases productivity, offering the comfort of a road vehicle during construction site and off-road missions.

Trakker Dakar

Iveco is also exhibiting the **Trakker**, driven by the champion Miki Biasion, which competed at the **2012 Dakar Rally**. Recently the vehicle won first place in another international race, the **29th Baja España-Aragón**, which is the sixth competition in the world championship Rallyes Tout Terrain.

THE IVECO STAND

The focus is on the heavy range

The stand covers over two thousand square meters and reflects colours and materials used for the creation of the Fiat Industrial Village in Turin. It presents some of the most representative vehicles of the different ranges and it is divided into sections, according to the ranges.

The spotlight at the stand centre is on the heavy range vehicles; right at the heart of the stand, there is an area dedicated to engines and engine technologies. All around the central area of the stand, the vehicles of the other ranges are exhibited, from light to medium-duty vehicles, including minibuses, buses and construction vehicles.

The entire exhibition area, further to be a products' showcase, shows how Iveco places the customer at the heart of its company strategy. From the design to the marketing of the product, from technical support to the definition of tailored financial services, attention is always focused on creating value for those who, on a daily basis, use Iveco vehicles for their work.

FPT Industrial innovative powertrain technologies in the heart of the stand

The engine technology area - in the heart of the stand - features **Euro VI engines** realised by **FPT Industrial**, the backbone of Fiat Industrial, which provides innovative engine solutions for all of the Group's companies. First with the Tier IV engines for CNH and then with the fully patented advanced catalytic reduction system **High Efficiency SCR (HI-eSCR)**, FPT Industrial

ensures absolute advantages over its competitors. FPT Industrial's after-treatment technology for Euro VI vehicles is unique and exclusive, since it meets the strict nitrogen oxide emissions limits through the catalytic reduction system, without any recirculation of exhaust gases.

At the exhibition area also light and medium range

Vehicles from the medium (Eurocargo) and light (Daily) ranges are also on show at the stand. In total, three **Daily** vehicles are exhibited, one of which is electric. This vehicle guarantees high reliability and is perfectly adaptable to ordinary urban missions, such as door-to-door distribution and urban passenger transport.

For medium range vehicles, the **Eurocargo**, best in class in Europe for medium haulage distribution, is shown in its Euro V EEV, as it is still the undisputed benchmark for the European market. In fact, it is a leader in the 6 to 19 t GVM segment.

Astra exhibits HD9 in Hanover

The off-road range also includes the **Astra brand**, spotlighted in Hanover with the new version of the **HD9**, an even more performing vehicle with a new cab. Astra represents an essential reference point in quarry/building-site transport for all big construction companies looking for a reliable partner, in terms of both strength and versatility of the vehicles and in terms of customer support.

Iveco collective passenger transport solutions on stage

Iveco presents the new **Daily Minibus** and the **Magelys Pro**. The new **Daily Minibus**, exhibited in the 170 HP **Daily Tourys version**, combines the traditionally recognised qualities of the Daily with the attributes of a leader in the collective passenger transport sector. The vehicle is the ideal solution for passenger transport operators for both urban routes and rapid extra-urban transit.

The **Magelys Pro** is represented by the 12-metre version, powered by a Cursor 10 EEV, with 450 HP and belongs to a vehicle family that in recent years has stepped up as a leader in the highly specialised Gran Turismo bus segment.

The outdoor area: test drives of the electric Daily

Iveco continues the exhibition of its products in an outside area too. These include four Daily, two Eurocargo, one of which powered by natural gas, two new Trakkers, two STRALIS Hi-Way and three other STRALIS Hi-Road, with medium cab, a Daily Minibus, a Crossway LE and a Leoncino. Visitors can test the Daily electric vehicle in the designated outside area.

The innovation area: "Iveco Dual Energy" chassis concept

Confirming Iveco's ongoing commitment to research into new technological solutions and possible future scenarios, there is **an area dedicated to innovation**, where the technology concept "**Iveco Dual Energy**", a chassis of a light-weight vehicle, a diesel-electric hybrid, is exhibited.

In the same area, Iveco unveils seats equipped with **Airbag "D-air enveloping airbags"**, developed by Dainese, to surround the human body with protection.

Aftersales and financing services

Finally, the attention to the customers is confirmed by the presence of an area entirely dedicated to after-sale services. A comfortable space, where visitors may be acknowledged and appreciate the wide offer of services and products assistance, original spare parts and accessories.

All Iveco after-sales products and services are conceived and continuously improved to ensure that the client benefits from unparalleled technical and financial assistance both at the time of purchase and throughout the entire life cycle of the vehicle. In the same area, **Iveco Capital** is present, in order to ensure customers maximum financial support.

IVECO INNOVATION TECHNOLOGIES FOR FUTURE MOBILITY

"Iveco Dual Energy": for tomorrow's light-duty vehicle

To meet the demands of increasingly sustainable mobility while responsibly combining economic growth with environmental protection, Iveco is committed to researching new technological solutions every day.

This innovative context led to the **Iveco Dual Energy**, a technological concept with a chassis that presents technology for the light-duty vehicle of tomorrow, shown in Iveco's innovation area in Hanover. Iveco Dual Energy proposes extremely flexible technology for light commercial vehicles, capable of switching to the most appropriate source of energy depending on the vehicle's mission.

The system relies on the use of two different types of traction - one is exclusively electric in order to ensure zero local emissions and low noise levels, and the other is hybrid (thermoelectric) and suitable for long journeys and for extra-urban missions, reducing consumption and CO₂ emissions by up to 25%. The principle of operation consists of just a few steps: the fuel's chemical energy is converted into mechanical energy for motion, whereas kinetic energy, which would be dispersed during braking or vehicle deceleration, is recovered and converted into electric energy to be stored in the battery. If necessary, this energy becomes available for the traction.

The hybrid system is the most appropriate for energy efficiency on extra-urban roads. Indeed, compared to a Diesel engine, this system ensures a reduction in fuel consumption and in CO₂ emissions of up to 25%. The electric system is indicated for urban or low-traffic areas, where the speed limit does not exceed 50 km/h. Then, alternating these two traction systems makes vehicles suitable for "last mile" management in zero emission areas, whilst ensuring the best performances for extra-urban activities.

Iveco worked on creating this concept using an open and collaborative means of innovation rooted in its ability to cooperate with important partners, sharing ideas, skills and enthusiasm with them. The chassis components on display are the result of teamwork that involved major international companies.

Specifically: the multi-function chassis was developed with ARCELOR MITTAL and MAGNETTO; the components made from fibre composite material with Xperion; the suspensions in multi-material with Streparava; the built-in plastic modules were developed in collaboration with Sole. Also, the e-drive, an electric engine-generator paired with an inverter, was developed in collaboration with Bosch; the multimode manual transmission with FPT Industrial; the lithium ion traction battery with Johnson Control; the

electric steering wheel with ZF Lenksysteme; the Electric Brake System with Brembo; the two-level air conditioning system with DENSO; the heat exchanger with CRF; the D-air® with enveloping airbags with Dainese.

In the same area, five seats are on display, four of which are for a Daily Minibus and one for a STRALIS, on which **Airbag "D-air enveloping airbags"** are installed, which use Dainese's D-air® technology.

D-air® is a technology that comes from the world of major motorcycle racing. Developed by Dainese, D-air® uses special high-pressure sacks with a patented 3D structure for maximum protection and ergonomics and "cold" technology gas generators. The new concept of a protective shell that envelops passengers was conceived to specifically address the needs of commercial vehicles and introduce a new paradigm in passive protection as well as a new way to improve the safety of transport.

The D-air® sack has a three-dimensional structure with controlled expansion (total intervention time from activation to inflation is around 45 milliseconds), which, together with high-pressure inflation, allows for a shape and volume designed to envelop and protect the passenger's body. All this makes it possible to achieve maximum shock absorption in a minimum amount of space - that is, the best possible trade-off between safety and ergonomics.

THE IVECO RANGE: TECHNICAL INFORMATION

STRALIS Hi-Way: working on the hi-way

Iveco presents the STRALIS Hi-Way, the latest generation of heavy-duty road vehicles. The new STRALIS Hi-Way, produced at the Iveco plant in Madrid, offers a series of important advantages to customers: reduced fuel consumption, lower maintenance costs and enhanced reliability and quality.

Hi-Profitability: the lowest cost of ownership over a vehicle life cycle

The needs of the customer are the most important factors for Iveco in addressing its product engineering solutions. With this in mind, Iveco developed the new STRALIS Hi-Way with the aim of reducing the **Total Cost of Ownership (TCO)**.

By combining product quality with a range of integrated services that include programmed assistance, fleet management and driver training for economical driving, **Iveco can obtain a TCO reduction by up to 4%** on an international mission carried out with a STRALIS Hi-Way tractor unit for an average of 130,000 kilometres a year and a holding period of 48 months.

As part of the costs that contribute to determining the value of the TCO, the most significant factor is represented by fuel consumption. Building on the ECOSTRALIS technologies already applied in the previous range, which ensured excellent fuel efficiency, the new STRALIS is equipped with next-generation solutions. In combination, these additions allow for a **savings of up to 10% in fuel**.

Hi-Design & Aerodynamics: the new front section

Designed with the primary aim of reducing drag coefficient (Cx), the new cab has been enhanced through new design features that render its unique style even more distinctive. The new front end is characterised by a central grille and redesigned air deflectors, a new sun visor with LED lights, optimized bumper design, light assemblies equipped with LED Daytime Running Lights and BiXenon headlights.

Aerodynamics is one of the most critical factors in determining energy efficiency: at 85 km/h, 40% of the fuel is used to overcome air resistance.

The new STRALIS cab is streamlined to achieve an excellent drag coefficient (Cx), improved by 3%. This makes the cab quieter for the driver, whilst delivering a significant fuel advantage on long-haul missions for the operator.

Hi-Comfort & Ergonomics: the new cabs

The STRALIS cab range offers two widths: the premium **Hi-Way** cab (width 2,500 mm), designed to ensure maximum comfort on long-haul missions for either single or two-man crews. This top-of-the-range cab is available in high and low-roof sleeper cab versions. Alternatively a medium cab (width 2,300 mm) is available as a sleeper cab **Hi-Road** with a choice of medium-high and low-roof, or as a day-cab **Hi-Street** low roof model.

All cabs have been redesigned around the driver, delivering exceptional ergonomics, comfort, safety and *infotainment*. This helps to support the creation of an optimal working environment where the focus is on maximising productivity.

Hi-Technology & Telematics: the on-board telematics

At the stand in Hanover, an area entirely dedicated to TELEMATICS will be located right next to the STRALIS range.

The pursuit of efficiency also makes use of advanced and easy-to-use control devices which help to provide a productive work environment. In particular, on the new STRALIS you will find IVECONNECT, the exclusive system with a 7" touchscreen that allows you to manage *infotainment*, navigation, driving support tools and advanced *fleet-management* services in an easy and integrated way.

IVECONNECT DRIVE includes a navigation system with truck navigation function, the "Driving Style Evaluation" system, which acts as a driving instructor always on board, and the Driver Attention Support safety device, to protect the driver from fatigue and drowsiness. This same *touchscreen* interface is also the on-board terminal for the IVECONNECT FLEET advanced fleet-management services, a new system that monitors the vehicle's position and allows for data integration with the company's logistics systems.

Hi-Efficiency: low fuel consumption and excellent performance

The new STRALIS is equipped with **Cursor FPT Industrial engines** with in-line six cylinder architecture, in three variants (8, 10 and 13- litres for the Euro V range and 9, 11 and 13- litres for the Euro VI range) and with power from 310 to 560 HP.

In Hanover, **Iveco makes two major announcements** on the engines available in STRALIS's Euro VI range. The range is enhanced by the powerful **Cursor 9 with 400 HP – 1700 Nm**, adding a competitive edge to fuel-transportation vehicles, as well as the **Cursor 11 with 480 HP – 2250 Nm**, which ensures a performance equal to that of a 13-litre engine but with the advantage of being lighter and consuming less.

They are engines with low fuel consumption and excellent performance, offering maximum torque at very low speeds (starting from 1,000 rev/min). Thanks to the reduced number of gear changes, this elasticity results in high efficiency and driving comfort. Exceptional vehicle operating range is guaranteed by fuel tanks with capacities of up to 1,400 litres. All powertrains are equipped with a decompression engine brake, which provides additional safety and helps to reduce the wear and maintenance costs of the braking system.

Furthermore, on all the Euro VI engines, there is an exhaust throttle valve that improves the performance of the new engine by up to 30% compared with current Euro V engines, called Super Engine Brake (SEB). The engine brake is integrated with the service braking system and with the auxiliary retarder Intarder.

Hi-Performance: the High Efficiency SCR system

The vehicle is characterised by highly innovative new engine technology, the **patented FPT Industrial system, Hi-eSCR**, which meets Euro VI regulations without compromising fuel efficiency. Thanks to the extreme efficiency of the HI-eSCR, the new STRALIS is the only heavy-duty vehicle on the market that meets Euro VI limits without using EGR (Exhaust Gas Recirculation).

With this exclusive solution, Iveco is able to simplify the after-treatment of gases and use the existing cab's "body in white", being that thanks to the HI-eSCR the engine does not require an increase in the capacity of heat exchange, and thus, does not need a greater radiant surface. Furthermore, the HI-eSCR system offers Iveco customers numerous other advantages such as weight reduction, lower fuel consumption and longevity due to less complex and more efficient technology.

Hi-Safety: safety is on board

The driver of a STRALIS can rely on an efficient and reliable braking system. The EBS now has the Brake Assistant function, for even more rapid and effective braking. The advanced safety systems are essential for protecting the driver, the vehicle and the load. The new STRALIS ensures safe driving for drivers and makes their work more relaxing and productive.

There are numerous safety systems available on the vehicle. Besides the already well-known **EBS (Electronic Braking System)**, **ESP (Electronic Stability Program)**, **ACC (Adaptive Cruise Control)** and **Hill Holder**, the **Lane Departure Warning System** is a system that warns the driver when the vehicle crosses the lines marking the traffic lane without the driver having turned on the turn indicators.

Moreover, the **Driver Attention Support** system constantly monitors the driver's level of attention by processing the movements of the steering wheel. The **Xenon headlights** increase the field of vision being lighted and improve the quality of lighting while the **Daytime Running Lights** guarantee maximum visibility of the vehicle.

Hi-Reliability: quality, high component reliability, low maintenance costs

The new STRALIS was developed by placing the highest possible emphasis on quality aspects. Quality controls during the manufacturing process were increased by 40% compared with the previous version of the vehicle. The new functions, the materials used, and the development processes have been subjected to the strictest quality controls, which extended to the creation of experimental fleets on which numerous reliability tests were undertaken. These included 6 million km of driving under particularly severe conditions, in order to verify the vehicle's

endurance, and 11 million km of road-reliability testing under normal load and driving conditions, some of which was performed by customers.

STRALIS Hi-Way “Emotional”

Visitors will also find the **STRALIS Hi-Way “Emotional”** on display at Hanover, a vehicle that offers a vast range of options, which integrate standard content with specific orderable content in customised packages according to the customer's needs. The STRALIS “Emotional” is fitted out with all of the new Iveco Parts&Services accessories designed to satisfy the driver's main necessities - both while driving and when stopped - and also to guarantee the driver's performance and safety as well as that of the load under any conditions.

Among the vehicle's main accessories are: the new Lavazza coffee machine for making espresso, a technologically advanced system that brings together high-quality sound and video, and a LED lighting system that creates an environment that favours both driving and relaxation.

In addition, Iveco's attention to sustainability issues is clearly shown by the option of having a photovoltaic panel installed on the roof hatch of the STRALIS “Emotional”, which allows batteries to be recharged even when the vehicle is stationary.

From an innovative perspective, with STRALIS Emotional Iveco focuses on the future, suggesting two innovative concepts, a possible evolution of the current range of optional available: **the Vehicle Control Center** and the **Full-LED Headlamps and Rear lamps Truck-lite**.

The **Vehicle Control Center** presents a new concept of user interface, based on a table app-based platform, whose design is especially tailored to enhance the system operability, making it both safer and easier to use while driving and in every other moment. The interface has been customised to give the idea of the Iveco family feeling and of a real integration of the object with the cabin space and the vehicle ambience.

The technology allows the access to a complete world of standard functionalities, ranging from streaming radio at user choice, web browser,

media player, socially interactive platforms: all of this will become part of the cabin experience, and when useful to the mission, will be used as facilitators of the driving task.

But in the interface presented there is much more: several customised apps especially designed for Iveco customers. This lean, innovative and smart interface will soon allow the user and the vehicle to communicate in the most efficient and intuitive way. Among these: demo interfaces for climate control; a simple tool for entering vehicle settings and memorize them linked to the user profile; a fleet management facility; an interactive version of the user's manual, that can be easily consulted; and a user-friendly way to display the diagnostics of the main components of the vehicle.

During the driving phase, the functionalities displayed in the interface will relate only to the vehicle's management, while during the vehicle stops all functionalities, including those related to web navigation, entertainment and socially interactive experience will be enabled.

The LED (Light Emitting Diode) products which may be used for the truck headlamps (parking light, low beam, high beam, Daytime Running Light) and rear lamps (stop light, parking light, fog light, reversing light, direction indicators) represent an innovative Iveco's proposal as a possible technological solution for the future. The advantages characterizing the use of this solution are: firstly long life, the a low current-draw lighting, compared to traditional incandescent lights (usually minus 80%) and an highest visibility throughput the overall nighttime use (+10% in longer field of vision, versus xenon lights).

STRALIS LNG Natural Power: alternative tractions for the transport business

With an ever keener eye for how to make its products environmentally friendly, Iveco dedicates a broad area to heavy-duty LNG-powered vehicles. Indeed, the stand includes a **Stralis LNG** (Liquefied Natural Gas), on display for the vast international public in Hanover.

In LNG vehicles the natural liquefied gas is stored as a liquid at -125°C in cryogenic tanks, and then fed to the engine as natural gas after being

heated in a heat exchanger. As for the reduction of emissions, compared with CNG vehicles, the LNG system also results in a reduction in unladen weight and greater range.

Furthermore, the use of natural gas-powered vehicles, whether CNG or LNG, represents a considerable economic asset for customers. In fact for regional distribution missions it has been estimated that savings of up to 10,000 Euros are possible compared with a traditional Diesel-powered vehicle, over a distance of approximately 40,000 km per year. On refuse collection and recycling missions it is estimated that the saving can be up to 20,000 Euros per year.

The market opportunities for these vehicles are particularly strong in food and beverage delivery, among logistics operators and firms engaged in night-time delivery, where the reduction of noise levels is guaranteed by spark-ignition engines. On average the noise reduction for gas fuelled Stralis vehicles is between 3 and 6 decibels compared to a vehicle with the same power rating running on diesel fuel.

The vehicle shown in Hanover is a 4x2 tractor unit built to transport 40 tons of GCW, with a Cursor 8 natural gas engine and 330 HP, manual gearbox, Intarder and EBS as standard. This version is ideal for regional and national missions, such as fuel transportation.

New Trakker: efficiency and value even off-road

Among the novelties, Iveco presents the new Trakker at the Hanover vehicle show. The vehicle, in its world premiere, has been renewed in its cab and in the telematics devices inside it. Reliability, efficiency, robustness and safety, these are the characteristics that make the new Trakker the ideal solution for everyday use as well as the toughest challenges.

Iveco's quarry and construction site vehicle can operate on impracticable, uneven terrain, withstanding constant stress with an incredibly powerful performance. In order to face heavy work, in terms of productivity for the customer, Iveco has come up with the perfect work vehicle, from its design to its robustness and functionality. This is a vehicle that raises productivity

beyond all imaginable limits, while offering the comfort of a road vehicle on construction-site and off-road missions.

Hi-Comfort & Ergonomics: the new cabs

The most important innovation was made to the cab, developed by borrowing from the solutions used in the heavy-duty road vehicles STRALIS, Hi-Street and Hi-Road. The new cab springs from the idea of enhancing driving comfort and time spent on board, factors already considered to be fundamental for productivity and safety.

Hi-Land is the short cab with low roof while **Hi-Track** is the long cab for missions that require the driver to spend the night on board and is available with either a low or a high roof. Entirely redesigned with mass-coloured top-quality plastics and anti-scratch treatments, the dashboard is highly ergonomic. All button and switch controls are easily visible and accessible while remaining perfectly seated.

Hi-Performance

The New Trakker is equipped with the Cursor engines which ensure power and endurance over time. The Trakker's Cursor engines - designed to guarantee excellent performance, low cost of ownership and superior driving comfort - are available in 8 or 13 litres and range from 310 to 500 HP. The Cursor 8, available in a 360 HP version with variable-geometry turbo, has displacement of 7,790 cm³ and in-line six-cylinder architecture, and meets EEV standards, which are even stricter than Euro V.

The Cursor 13 is ideal for the most demanding applications in the toughest environments and most adverse conditions, delivering high torque at low rpm for great drivability. The Cursor 13 is available in the 450- and 500- HP versions with variable-geometry turbo and in the 410-HP version with fixed-geometry turbo with a waste-gate. The New Trakker may also be equipped with an Intarder to further reduce the consumption of friction material by the brakes.

Hi-Reliability

The main feature of the new Trakker is its robustness. Starting with the steel chassis with a high yield limit, each individual component guarantees performance that endures over time. The ZF gearboxes available on the

new Trakker are all about ergonomics and performance. For example, the manual 9-speed and 16-speed Ecosplit gearboxes are equipped with a servo-shift system for safer, more comfortable driving and the automatic 12-speed and 16-speed gearboxes with a gear selector built into the steering column stalk, which makes driving more comfortable.

Astra HD9 facing the international audience

The **Astra** brand is also under the spotlight with presentation in Hanover of the new version of the **HD9**, with the exhibition of the 8x4 84.44 model.

The characteristics of these versatile yet robust vehicles and the countless customization options make it possible to set up the in such a way as to satisfy any transportation and site requirement, in any part of the world, from the quarry to the construction sector, from special haulage to the Oil & Gas transportation sector. Today, with the new **HD9** series, Astra is once again the leader on the world stage with an even more performing vehicle with a new cab.

The newly designed HD9 cab best encapsulates the confident, unique personality of Astra vehicles. The cab is built out of processed steel and has a new and aggressive design. The new grille layout, which is very visible thanks to its rounded lines, can open up entirely with the connected corner elements allowing easy access to the maintenance areas.

The interior, too, has been completely renovated in terms of functional elements and fittings and includes solutions that improve living and driving comfort. Access to the cab is made easier by the ideal spacing of the first two steps and the double vertical railing on both sides of the door, as well as the fact that the door itself opens up by more than 90°.

The entire kinematic chain is designed to handle the most challenging missions: thanks to the exceptional torque specifications at the different engine speeds, the 13 liter Cursor engines with electronically - controlled pump injectors and double overhead camshaft, guarantee the HD9 highly versatile on-road performance and exceptional braking power - essential features when operating on a construction site.

Astra also offers a wide range of choice of mechanical gearboxes from those with Ecosplit 4 technology (ZF with 16 speeds) to automated Astronic 16-speed gearboxes to Allison automatic gearboxes, for a maximum engine performance in terms of peak power and excellent traction on all types of terrain.

Depending on the load, the Astra offers the option of equipping its vehicles with front axles and motor axles bearing 8t, 8.5t, 9t loads and even offers a reinforced version with a load bearing capacity of 10t on 6x6 motorized axle models, designed for particularly tough applications, on shifty terrain such as sand or mud; or with special set ups with loads concentrated on the front axle such as tow systems, cranes or boring structures.

The real strong point of the Astra vehicles is its chassis, built out of special steel with the highest level of resistance and high elasticity, thanks to the two flat beams running parallel to each other for the entire length with a C section (320 x 90 x 10mm) and joined to each other by crosspieces. With the highest Rail Bending Moment (R.B.M.) compared to competitors' vehicles, Astra's chassis is renowned for its ability to transport heavy loads on any terrain conditions, reducing torsion stress and ensuring great stability even with very high centers of gravity.

Daily: the light vehicle suitable for every kind of work

Powerful, environmentally friendly and reliable, the **Daily** showcases all its features at Hanover: light and versatile so as to meet the transport demands of any professional.

Its truck-style chassis frame, rear-wheel drive, direct injection turbocharged diesel engine, 4x4 version, intercooler, natural-gas engines, up to 17.2 m³ of loading space volume and 210 cm internal height, common rail fuel injection, six-speed transmissions and seven tons maximum permitted mass. In brief, these are the strengths that summarise the history of an undisputed star in the light-duty commercial vehicle market.

The Daily offers the world's most innovative engines in the world of commercial transport: the most powerful diesel engine in its segment, the Euro 5, 3-litre Twin Turbo, rated at 205 HP (150 kW) with a maximum

torque of 470 Nm; the new Euro 5, 2.3-litre diesel engine at 146 HP (107 kW) with a torque of 350 Nm, with a new variable-geometry turbocharger and the exclusive Multijet II fuel-injection technology, optimised to provide maximum operating efficiency; a new EEV version of the 3-litre diesel engine, rated at 146 HP and 370 Nm with a variable-geometry turbocharger completes the new engine range, offering tailpipe emissions of particulate matter well below the levels required by the homologation standards.

The range also features the Natural Power Bi-fuel version, powered by the super-ecological EEV engine and optimised to run on natural gas, with a small gasoline tank for use in case of emergency. It includes the zero-emission Daily electric as well, designed, produced and sold directly by Iveco.

But that's not all, because, as well as enjoying the benefits of these new engines, the driver can improve and optimise his driving style with new technologies like the Stop & Start functionality and the GSI (Gear Shift Indicator) system, enabling the driver to better manage his way of driving. Together with the new six-speed transmission offered with 2.3-litre diesel engines, these functions reduce fuel consumption, and consequently, CO₂ emissions of up to 10% compared with the previous model.

Iveco has always distinguished itself for its attention to road safety. The new Daily sees the introduction of the latest generation ESP 9 system, featuring active sensors, including ABS, EBD, ASR (traction control), MSR (over-run torque limitation), Hill Holder (electronic system for starting the car while uphill), LAC (Active load control), as well as new features such as the HFC (Hydraulic brake fade control), RMI (Roll movement intervention), ROM (Roll over mitigation) and TSM (Selective braking enhancing vehicle and trailer dynamic stability). This guarantees that the vehicle can be driven as safely as possible.

Eurocargo: an ideal solution for any challenge

In a constantly evolving work world, one has to be able to adapt with maximum versatility in order to stay competitive. With **Eurocargo** Iveco

offers its clients a vehicle that comes in a myriad of different combinations, thanks to the numerous variants in the range and the option to choose your own suspensions, axles and final drive ratios most suited to the specific mission. It has the richest and most versatile range in its category, with 14 variants of gross vehicle weight (from 6 to 19 t), seven horsepower categories (from 140 to 300 HP), 13 gears (seven manual, three automated and three automatic), 4x2 and 4x4 drive, 13 wheel bases (from 2,790 to 6,570 mm) and three types of cabs with two roof heights. The options in the range vary from bodies to tarpaulins, from isothermal trucks to three-sided tippers, from cisterns to shops on wheels, from loading platforms to aerial platforms to fire-fighting fittings and even down to small concrete mixers and road sweepers.

Eurocargo's flexibility and robust structure is enhanced even more in the Tector engine range. The secret of Eurocargo's Tector engines is their elasticity; the torque values, which are high and constant for a broad range of revolutions, guarantee smooth driving, fewer gears, less wear on the engine and greater energy efficiency. All Eurocargo engines are also EEV (Enhanced Environmentally Friendly Vehicles) approved.

Eurocargo is also available specifically powered by methane, CNG (Compressed Natural Gas), with gross vehicle weight from 11 to 16 t and a six-cylinder engine with 200 HP.

The vehicle has proven itself to be ideal for driving in the city, where it manoeuvres incredibly well, and also on mixed routes and motorways. Also from the standpoint of traveller, driver and pedestrian safety as well as from that of the load transported and the investment value, the vehicle delivers top performance.

The vehicle is available with three different types of cab: short (MLC), ideal for distribution and short routes; long with a low or high roof (MLL) for long hauls and sleeping on board; double (MLD) for six people plus the driver.

Passenger transportation: from minibuses to luxury buses

Iveco has always been true to its mission as a builder sensitive to the issues of ever more sustainable, functional and ecological mobility. The company is one of the main distributors of hybrid vehicles in Europe, and with the hybrid vehicle ranges Citelis and Access'Bus GX in the 12- and 18-metre versions, it is able to offer efficient vehicles in terms of comfort, energy savings and emissions reduction. The environmental impact is highly satisfying; with the hybrid system the average reduction in fossil-fuel consumption and CO₂ emissions ranges from 25 to 35%.

Lowering CO₂ emissions by 35% translates into approximately 500 grams less of CO₂ per km. Considering an annual distance travelled of 50,000 km (which is the average value for a bus), it is possible to cut approximately 25 tons of CO₂ from being emitted into the atmosphere.

At Iveco's stand in Hanover, there will also be vehicles from the passenger transportation range. Demonstrating the complete offering provided by Iveco, which spans from minibuses to Gran Turismo buses, the company shows the **new Daily Minibus and the Magelys Pro**.

The Daily Minibus range includes more than one hundred configurations for the transport of up to 30 passengers in five different versions and may be used for various missions, from urban to scholastic transportation to tourism. Well-being on board is a top priority for the Daily Minibus. The large side window ensures that the vehicle is extremely well-lit and offers a panoramic view of the surrounding landscape. The ergonomic driver station is even more improved and includes the features of the new model.

The standard outfitting includes ABS (Antilock Braking System), ASR (Acceleration Slip Regulation), ESP (Electronic Stability Program), Hill Holder (electronic system for starting the car while uphill) and fog-lights with Fog Cornering function. These systems make the new Daily minibus one of the safest vehicles in its category. In addition, an airbag for the driver, speed limiter and an electric brake may be added to the vehicle.

The joy of driving and respecting the environment are the main characteristics of the FPT Industrial three-litre engine. Available in two diesel EEV versions (146 HP or 170 HP) and one CNG version (136 HP),

the engine is characterised by a maximum torque of 400 Nm at 1,250 revolutions/min (for the 170 HP engine), low emissions levels, reduced fuel consumption, as well as power and high elasticity.

To complete the picture for this sector, a **Magelys Pro** is on display, a new version of a family of vehicles that in recent years has earned a leading position in the highly specialised segment of Gran Turismo buses. With top performance, optimised fuel consumption, low environmental impact, long-lasting reliability, and above all, outstanding driver and passenger comfort, it possesses winning characteristics that have hit the mark when it comes to customer expectations.

The Magelys Pro range is available in 12.2 and 12.8 m versions with flatbed height HD. The Magelys Pro has a one-piece rigid steel body that has undergone cathaphoretic treatment against corrosion and also exploits the most advanced technologies to aid the driver in his daily tasks. Braking has been improved with the electronic EBS system, the ESP system that regulates vehicle stability, and ACC (Adaptive Cruise Control), available as an option, which ensures that distance from the vehicle ahead remains constant.

The choice of a tested and proven kinematic chain in the Cursor 10 engine, Euro V, of 380 or 450 HP, paired with six-speed mechanical gearboxes or 12-speed robotic gearboxes, guarantees the reliability of this vehicle made to last over time.

Inside area: the vehicles exhibited

1	DAILY	35S15V S&S 12m3
2	DAILY	70C21 w3450
3	DAILY	35SElectric 12m3
4	EUROCARGO	ML180E30/P
5	STRALIS	AS440S50T/P (emotional concept)
6	STRALIS	AT440S33T/P LNG
7	STRALIS	AS440S46T/P
8	STRALIS	AS260S46Y/FS-CM
9	STRALIS	AS 440S46TX/P
10	STRALIS	AS440S50T/P
11	TRAKKER (LAUNCH)	AD410T50P
12	TRAKKER DAKAR (FIV)	AT190T44W
13	MINIBUS DAILY	Daily Tourys ERO
14	MAGELYS	PRO
15	ASTRA	HD9 8x4

Outside area: the vehicles exhibited

1	DAILY	70C17K
2	DAILY	35S17D
3	DAILY	55S17W
4	DAILY	35S15V
5	EUROCARGO	75E18P
6	EUROCARGO	120EL20P-CNG
7	STRALIS	AS260S46Y/PS
8	STRALIS	AT260S42Y/PS
9	EUROCARGO	160E30
10	STRALIS	AT440S40T/P
11	STRALIS	AT440S42T/P-SL
12	STRALIS	AS440S50T/P
13	TRAKKER	AD410T50
14	TRAKKER	AD260T45
15	CROSSWAY	LE CITY
16	LEONCINO	



Iveco

Iveco, a Fiat Industrial Group company, designs, manufactures and markets a wide range of light, medium and heavy commercial vehicles, quarry/construction site vehicles, city and intercity buses and special vehicles for applications such as firefighting, off-road missions, defence and civil protection.

With approximately 25,000 employees, Iveco manufactures in 11 countries around the world using the most advanced technologies. The company also operates in Europe, China, Russia, Australia and Latin America. Approximately 5,000 sales and service points in over 160 Countries guarantee support in any geographic area where there is an Iveco vehicle at work.

Hanover, 18th September 2012